

Lab 1 - Requirements Table

Problem Statement #19: Personalized Meal & Diet Subscription Manager (Healthcare & Telemedicine)

Actors: Subscriber, Dietitian (Payment Gateway as external system actor)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall generate weekly meal plans strictly adhering to the subscriber's macro-nutrient constraints and allergen exclusion list.	High	Pass: Generated menu contains zero excluded allergen ingredients and stays within the subscriber's daily calorie/macro range. Fail: Any excluded allergen (e.g., peanuts) appears in a plan for a peanut-allergic profile.	Directly protects patient safety; an allergen breach can cause serious clinical harm.
FR-002	Functional	The system shall require a registered Dietitian to review and approve every auto-generated weekly meal plan before it is released to the subscriber.	High	Pass: A plan status changes from 'Draft' to 'Released' only after a Dietitian records an 'Approved' action against it. Fail: A plan reaches 'Released' status with no linked Dietitian approval record.	Clinical oversight is required for a diet plan tied to medical constraints such as diabetic limits.
FR-003	Functional	The system shall allow a subscriber to update their dietary profile (allergies, diabetic limits, calorie goals) at any time, and shall regenerate the next pending meal plan using the updated profile.	Medium	Pass: After a profile edit is saved, the next not-yet-dispatched plan reflects the new constraints within 1 menu-generation cycle. Fail: A pending plan still reflects the pre-edit profile after the update is saved.	Patient conditions (e.g., a new allergy) can change and must be reflected without delay.
FR-004	Functional	The system shall record and display the daily delivery fulfillment status (Delivered, Missed, Rescheduled) for each subscriber's meal order.	Medium	Pass: Every dispatched order has exactly one fulfillment status visible on the subscriber's dashboard within 30 minutes of the delivery window closing. Fail: An order remains with no status 1 hour after its delivery window closes.	Enables the subscriber and dietitian to trust that clinically planned meals are actually being consumed as scheduled.
FR-005	Functional	The system shall send a reminder notification (email or SMS) to the subscriber 24 hours before each scheduled meal dispatch.	Low	Pass: A notification log entry exists for every dispatch, timestamped between 23-25 hours before the dispatch time. Fail: A dispatch occurs with no corresponding reminder logged.	Gives the subscriber a final window to pause or reschedule, reducing food waste and missed deliveries.
NFR-001	Nonfunctional (Performance)	The subscription engine must support schedule modifications (pause, skip, reschedule) up to 12 hours before dispatch.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load; a modification submitted at the 12-hour cutoff is accepted and reflected before dispatch. Fail: A modification within the allowed window is rejected or not reflected before dispatch.	Kitchens/logistics need a firm cutoff to re-plan dispatch batches without disrupting other orders.
NFR-002	Nonfunctional (Security &	The system shall encrypt all patient dietary and health-constraint data (allergies, diabetic limits)	High	Pass: Security scan/audit confirms encryption at rest and in transit on all health-data stores	Dietary/allergy data is sensitive health information; protecting it is a clinical and

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	Privacy)	at rest (AES-256) and in transit (TLS 1.2+), and restrict access to authenticated Subscriber and Dietitian roles only.		and endpoints, and unauthorized role access attempts are logged and blocked. Fail: Any health-data endpoint is reachable unencrypted or by an unauthenticated/unauthorized role.	regulatory requirement.